INCH-POUND
MIL-DTL-24640/10E
26 June 2008
SUPERSEDING
MIL-C-24640/10D
22 August 2002

DETAIL SPECIFICATION SHEET

CABLE, ELECTRIC, 600 VOLTS, LIGHT-WEIGHT, MULTI-CONDUCTOR, AWG 16, TYPE MXSO AND MXSOW

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24640.

Construction, Non-Watertight (MXSO) and Watertight (MXSOW)

First	-	16AWG, tin-coated copper conductor. Non-watertight constructions shall be constructed using 19 strands. Watertight constructions shall be constructed with 7 strands.
Second	-	Insulation material and basic construction shall be in accordance with SAE-AS81044/12, and shall be tested in accordance with the component wire ratings and requirements listed on this specification sheet. Conformance testing of components shall be in accordance with SAE-AS81044, prior to cabling. Identification in accordance with Method 1, component wire shall be white.
Third	-	The required number of singles shall be cabled together. Cabling sequence shall be consecutive, starting with number 1 from center outward. Fillers may be used as necessary, to form a round cable.
Fourth	-	Binder tape applied helically with overlap.
Fifth	-	Braided shield, tin-coated copper strands. Braid must be constructed to meet surface transfer impedance requirements as listed herein.
Sixth	-	Binder tape applied helically with overlap, at manufacturer's option.
Seventh	-	Cross-linked polyolefin jacket. (See table I for thickness.)

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TABLE I. Construction details.

PIN M24640/10	Type Number	Number of Component Wires	Diameter Over Shield (inch) (nom.)	Jacket Thickness (inch) (min.)	Cable Diameter (inch) (min.)	Cable Diameter (inch) (max.)	Cable Weight (pounds per 1000 feet) (max.)
-01UO	MXSO-2	2	0.190	0.034	0.282	0.304	76.4
-02UO	MXSO-9	9	0.310	0.041	0.424	0.458	180.0
-03UO	MXSO-21	21	0.425	0.044	0.552	0.596	329.0
-04UO	MXSO-37	37	0.540	0.044	0.644	0.694	497.0
-05UO	MXSOW-2	2	0.200	0.034	0.282	0.304	83
-06UO	MXSOW-9	9	0.320	0.041	0.424	0.458	200.0
-07UO	MXSOW-21	21	0.435	0.044	0.552	0.596	365.0
-08UO	MXSOW-37	37	0.550	0.044	0.644	0.694	550.0

REQUIREMENTS:

Qualification required.

Construction details — See table I

Component wire — The component wire shall be in accordance with SAE-AS81044/12 and the following:

- Temperature rating — 150 °C (302 °F) (max.) conductor temperature
- Voltage rating — 600 volts (root mean square) at sea level
- Finished component diameter
  - Non-watertight — 0.072±0.003 inch
  - Watertight — 0.073±0.003 inch
- Component wire examination and tests — The component wire shall be tested in accordance with the methods listed in SAE-AS81044 and SAE-AS81044/12 and the following requirements:
  - Accelerated aging — Oven temperature, 250±2 °C (482±3.6 °F) for 0.6 hours
  - Low temperature cold bend — -40 °C (-40 °F) for 4 hours
  - Blocking — 150±2 °C (302±3.6 °F)
  - Flammability — 30 seconds (max.); 3.0 inches (max.); no flaming of tissue paper
  - Humidity resistance — 5000 megohms for 1000 feet (min.) insulation resistance after humidity exposure
  - Impulse dielectric test
  - Primary insulation (when tested in lieu of spark test)
    - 6.0 kilovolts (peak) 100 percent test
    - Finished wire — 8.0 kilovolts (peak), or 3 kilohertz spark test

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5.5 kilovolts, 100 percent test	
Insulation resistance	5000 megohms for 1000 feet (min.)
Life cycle	Oven temperature, 200±2 °C (392±3.6 °F) for 120 hours
Physical properties of primary insulation	Tensile strength, 2500 pounds per square inch (psi) (min.) Elongation, 150 percent (min.)
Shrinkage	0.125 inch (max.) at 250±2 °C (482±3.6 °F)
Spark test primary insulation	1500 volts (root mean square), 60 hertz or 3 kilohertz, 100 percent test
Surface resistance	500 megohms inches (min.), initial and final readings
Thermal shock	Oven temperature, 150±2 °C (302±3.6 °F)
Maximum change in measurement	0.060 inch
Wet dielectric test	2500 volts (root mean square)
Wrap test	Mandrel wrap test required

EXAMINATION AND TESTS:

Basic Electricals:	
Conductor resistance (ohms/1000 feet at 25 °C, max.)	5.15
Voltage withstand (volts, root mean square, min.)	
Conductor to conductor	2000
Conductor to shield	1000
Shield to shield	Not Applicable
Insulation resistance (megohms/1000 feet, min.)	2500
Conductor and shield continuity	No Failure
Jacket flaws	No Failure
Group A:	
Visual and dimensional	No Failure
Capacitance:	
Mutual capacitance (at 1 megahertz, picofarads/foot, max.)	Not Applicable
Capacitance unbalance (percent, max.)	Not Applicable
Characteristic impedance (at 1 megahertz, ohms)	Not Applicable
Watertightness (Type MXSOW only) (see table IX of MIL-DTL-24640 for limits of water leakage)	No Failure
Weight	No Failure

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Group B:	
Cold bending	No Failure
Drip	Not Applicable
Cross-linked proof test (percent, max.)	
Jacket	50
Tear (pounds per inch thickness, min.)	35
Physicals (unaged)	
Jacket	
Tensile strength (lb/in^2, min.)	1300
Elongation (percent, min.)	160
Attenuation (at 3megahertz, decibels/100 feet, max.)	Not Applicable
Gas flame (1 hour)	Not Applicable
Group C:	
Physicals (aged)	
Jacket, air oven	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Permanence of printing [(conductor) (Method 1 only)] (cycles, min.)	25
Permanence of printing (jacket) (cycles, min.)	125
Shield (conformance to material, construction, and coverage)	No Failure
Heat distortion (percent of unaged, max.)	30
Cable sealant removability (Type MXSOW only)	No Failure
Shrinkage	No Failure
Mutual inductance (decibels/1000 feet, max.)	Not Applicable
Surface transfer impedance	
Milliohms per meter (max.)	700
EMP response (decibels, min.)	60
Pulse response time (milliseconds/500 feet)	
Rise time, max.	Not Applicable
Decay time, max.	Not Applicable
Qualification Inspection:	
Qualification inspection shall include basic electricals, groups A, B, and C, plus the following:	
Flame propagation	No Failure

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Aging and compatibility (cable)	No Failure
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Insulation	0.2
Hot oil immersion	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index, max.	
Jacket	25
Fillers	45
Insulation	10
Toxicity index, max.	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) [(Type A) hardness, min.]	80
Weathering (jacket)	No Failure
Fungus resistance	No Failure
Abrasion resistance	No Failure
Gas flame (3 hour)	Not Applicable
Component wire	No Failure
UNIT ORDERING LENGTHS: 2 & 9:	500 feet nominal
21 & 37:	250 feet nominal

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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Custodians:
Army – MI
Navy – SH

Preparing Activity:
Navy – SH
(Project 6145-2008-100)

Review Activities:
Army – AV, CR, CR4
Navy – CG, EC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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